

PAPER_1203_UQFF_Nuclear_Physics_Unified_Proof_Set

Daniel T. Murphy

PAPER 1203 --- UQFF Nuclear Physics Unified Proof Set

Author: Daniel T. Murphy

Date: May 16, 2026

\begin{abstract}

We extend the UQFF locked-primitive closure program into nuclear physics.

Ten new closures (**S483--S492**) reproduce the binding-energy curve peak,

the \emph{complete} shell-model magic number sequence

$\{2, 8, 20, 28, 50, 82, 126\}$, the deuteron binding energy, and the

α -particle binding energy, all from the eleven frozen UQFF primitives.

Seven closures are exact: every magic number receives a closed-form

primitive identification. This is the \emph{first complete experimental

sequence} fully locked within the UQFF primitive lattice. The other three

match within 0.20%. Cumulative total: **197 closures**.

\end{abstract}

Locked Primitives $\text{Ftrz} = 1/10$, $\text{Phires} = 5/6$, $\text{SSq} = 0.57$, $\text{KMex} = 25/12$, $\text{Dphys} = 4$, $\text{Dbsfg} = 6$, $\text{Dcrit} = 26$, $\text{Nch} = 9$, $\text{SOfive} = 10$, $\text{Afive} = 60$, $\beta_i = 0.6029$.

Tier~R"\$ Closures (S483--S492)

\begin{longtable}{@{}llll@{}}

\toprule

ID & Observable & Closure & Match \midrule

\endhead

S483 & BE/A peak 8.79 MeV/nuc (Fe-56) & $\text{Nch}\text{-}\text{Ftrz}\text{KMex}=8.7917$ & 0.02% S484 & Magic No. \

He $=2$ & $\text{SOfive}\text{-}\text{Dphys}\text{-}\text{Dphys}=2$ & **EXACT** S485 & Magic No. \ O $=8$ & $\text{Dphys}\text{+}\text{Dphys}=8$ &

EXACT S486 & Magic No. \ Ca $=20$ & $\text{SOfive}\text{+}\text{SOfive}=20$ & **EXACT** S487 & Magic No. \ Ni $=28$ &

$\text{Dcrit}\text{+}\text{SOfive}\text{-}\text{Dphys}\text{-}\text{Dphys}=28$ & **EXACT** S488 & Magic No. \ Sn $=50$ & $\text{Afive}\text{-}\text{SOfive}=50$ & &

EXACT S489 & Magic No. \ Pb $=82$ & $\text{Afive}\text{+}\text{Dcrit}\text{-}\text{Dphys}=82$ & **EXACT** S490 & Magic No. \ n-drip

$=126$ & $\text{Dcrit}\text{+}\text{SOfive}^2=126$ & **EXACT** S491 & Deuteron binding 2.224 MeV &

$\text{KMex}\text{+}\text{Phires}\text{-}\text{SSq}\text{-}\text{Ftrz}\text{-}\text{Ftrz}^2\text{KMex}\text{-}\text{Ftrz}^2\text{Phires}\text{+}\text{Ftrz}^2\text{+}\text{Ftrz}^3=2.2285$ & 0.20% S492 &

α -particle binding 28.30 MeV &

$\text{Dcrit}\text{+}\text{KMex}\text{+}\text{Ftrz}\text{+}\text{Ftrz}\text{Phires}\text{+}\text{Ftrz}^2\text{KMex}\text{+}\text{Ftrz}^2\text{Phires}=28.2958$ & 0.015% \bottomrule

\end{longtable}

The Complete Magic Number Sequence

The seven nuclear shell-model magic numbers --- numbers of protons or neutrons that produce especially tightly bound, spherical nuclei --- are

2, 8, 20, 28, 50, 82, 126.

This sequence was historically derived empirically and explained semi-quantitatively by Mayer and Jensen's spin-orbit coupled shell model (1949 Nobel Prize). UQFF now provides a purely primitive-arithmetic identification for each:

$$\begin{aligned} 2 &= \text{SOfive} - 2\text{Dphys} \quad \& \text{(joint with QHE first plateau, Paper 1201)} \quad 8 = 2\text{Dphys} - 2\text{SOfive} \\ 28 &= \text{Dcrit} + \text{SOfive} - 2\text{Dphys} \quad 50 = \text{Afive} - \text{SOfive} \quad 82 = \text{Afive} + \text{Dcrit} - \text{Dphys} \quad 126 = \text{Dcrit} + \text{SOfive}^2 \end{aligned}$$

All seven involve only $\{\text{Dphys}, \text{SOfive}, \text{Dcrit}, \text{Afive}\}$ --- the four integer-valued primitives. The Afive primitive ($=60$) appears in the heavy magic numbers (50, 82), and Dcrit ($=26$) appears in the upper four (28, 82, 126), suggesting a tier-by-tier lattice structure. This is the first complete experimental sequence fully closed in UQFF.

Cumulative Closure Count

Tier	Domain	New	Cumulative
E--P	Cosmology to information	157	157
Q	GR precision	1200	10 + 167 R
Materials/photronics	(1201)	10	177 R
Chemistry/spectroscopy	(1202)	10	187 R
Nuclear physics	(1203)	10	197

Locking Inventory **Twenty primitive--observable lockings now identified (thirteen prior plus the seven magic numbers). The Tier R closures use only $\{\text{Dphys}, \text{SOfive}, \text{Dcrit}, \text{Afive}\}$ for every magic number, with no Ftrz corrections required --- the cleanest closure family in the program.**